HSBI

DUNGEON

Hochschule Bielefeld
University of Applied Sciences and Arts

Designing Workshops to Bridge Visual and Textual Programming

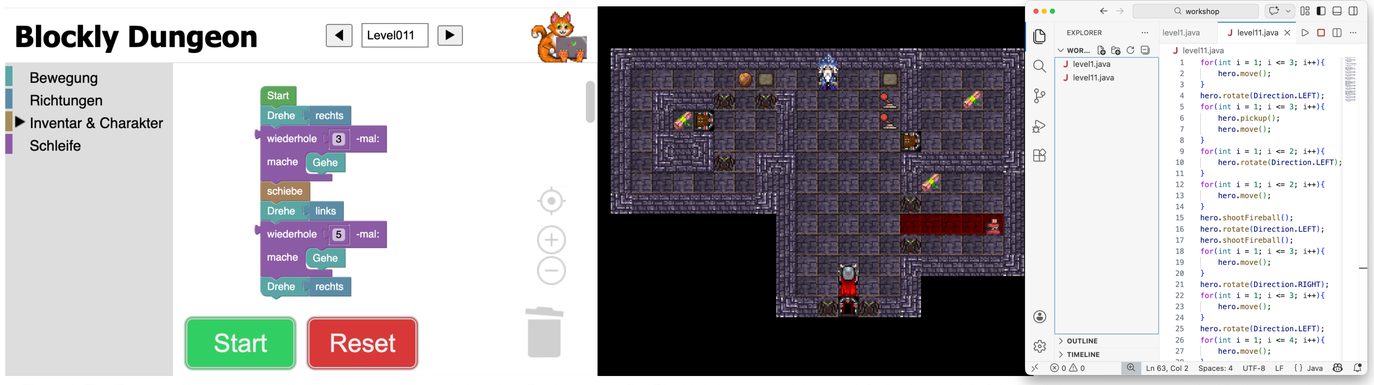


Figure 1: The Dungeon Game (center) displays the current puzzle to solve. Beginners use the Blockly interface (left) to complete the level, while intermediate programmers can tackle the same level using Java code in the Visual Studio IDE. Both input methods can be switched back and forth, enabling a seamless transition between visual and textual programming.

The Bridge

Schools teach visual block-based programming, yet universities and industry demand textual languages such as Java. This transition is a well-documented hurdle that often discourages learners [1].
Most approaches treat the two paradigms as separate courses and leaving learners to jump across the gap on their own.
We built the bridge instead. The **Dungeon Workshops** are a multi-level workshop concept centered on an 2D dungeon crawler developed at HSBI.

The key idea: the same narrative game world is accessible through both a visual and textual Interface (Fig. 1). Learners move from one paradigm to the next without changing the task context, the story, or the game only the tool in their hands changes. This continuity provides a coherent and motivating learning progression across different programming stages [2], grounded in constructivist principles [3].
Progressively challenging quests keep learners engaged while demanding more expressive programming over time - **from snapping blocks to writing code**.

Workshops

Three workshop formats are offered to address different prior knowledge levels:

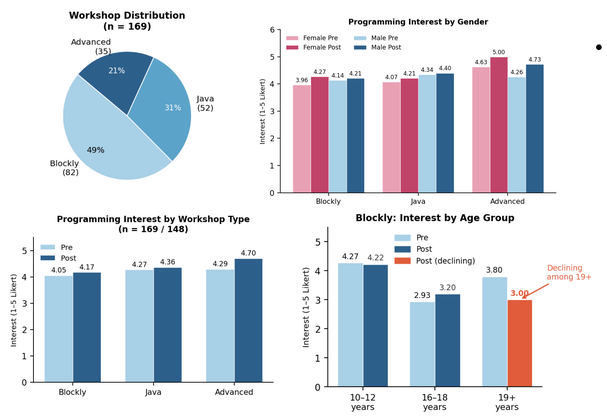
Blockly Dungeon: Aimed at pupils from grade 5 onward with no prior programming experience. Participants use a visual block-based editor. They program hero behaviour to solve puzzles, and observe immediate in-game feedback. The goal is to build algorithmic thinking and demystify programming through play.

Java Dungeon: Designed for learners who have some programming background. Participants write Java code to control the hero and solve puzzles. A custom VS Code extension provides scaffolded code navigation and live feedback, easing the transition from visual to textual programming

Advanced Dungeon: Targeted at experienced pupils. Participants implement complex game mechanics inside the game engine itself. The focus shifts from guided exercises to open-ended problem-solving within a realistic software project

Results

12 workshops were conducted with **169 participants** (grades 5-13). Post-workshop surveys were completed by 148 participants. Programming interest was measured on a 5-point Likert scale before and after each workshop. All three workshop types showed increased interest in programming after participation.



Limitations

Results should be interpreted with caution. The survey captures **self-reported interest** only - two important dimensions remain unmeasured:

A. **actual knowledge gain**, as no pre/post skill assessment was conducted, and
B. **long-term effects,** as follow-up data on whether interest persisted after the workshops was not collected.

Sub-group sample sizes are small. No control group was used, and data collection is still ongoing**.**

Future Work

- Expand the evidence base**: A further workshop phase and a hackathon are already planned to gather more data and stress-test the formats with a broader audience.
- Lower the barrier to adoption: All workshop materials are already available as OER OSS. Our next focus is simplifying onboarding so that other institutions can run the workshops with minimal setup effort.
- Grow the community: We invite educators and developers to reuse, adapt, and contribute back to the framework and workshop materials.

Literatur
[1] Weintrop, D. & Wilensky, U. (2017). Comparing Block-Based and Text-Based Programming in High School CS. *ACM Trans. on Computing Education, 18*(1).
[2] Alshahrani, A. et al. (2021). Using games to teach computer programming. **Computers in Human Behavior Reports, 3*.
[3] Petri, G. & Gresse von Wangenheim, C. (2017). How games for computing education are evaluated? *Computers and Education, 107*, 68–90.

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Dungeon on GitHub

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